Question no. 1

$z = -5 + 8i$ **then find** $z - z^*$

Solution:

$$z = -5 + 8i$$
$$z^* = -5 - 8i$$
$$z - z^* = -5 + 8i - (-5 - 8i)$$
$$= -5 + 8i + 5 + 8i = 16i$$

Question no. 2

If $z = 3 + 6i$ **then find** $z.z^*$

Solution:

$$z = 3 + 6i$$
$$z^* = 3 - 6i$$
$$z.z^* = (3 + 6i)(3 - 6i)$$
$$= 9 - 18i + 18i + 36 = 45$$

Question no. 3

Evaluate

$$\left| \frac{4 + 7i}{-5 + 6i} \right|$$

Solution:

$$\frac{4 + 7i}{-5 + 6i} * \frac{-5 - 6i}{-5 - 6i} = \frac{(4 + 7i)(-5 - 6\mathrm{i})}{(-5 + 6i)(-5 - 6i)}$$
$$= \frac{-20 - 24i - 35i - 42i^2}{(-5)^2 - (6i)^2}$$
$$= \frac{22 - 59i}{25 + 36} = \frac{22 - 59i}{61} = \frac{22}{61} - \frac{59}{61} i$$
$$\left| \frac{4 + 7i}{-5 + 6i} \right| = \sqrt{(\frac{22}{61})^2 + (\frac{59}{61})^2} = \sqrt{\frac{484 + 3481}{3721}} = \sqrt{\frac{3965}{3721}}$$

Question no. 4

Write the domain and range of $f(x) = 6x + 9$

Solution:

Domain of f(x)= All real no's

Range of f(x)= All real no's

Question no. 5

Write the domain and range of $g(x) = \frac{5}{x+6}$

Solution:

Domain of g(x)= $R - \{-6\}$

Range of f(x)= All real no's except zero

Question no. 6

Prove that $f(x) = x^4 + 5x^2 + 3$ is an even function

Solution:

$$f(x) = x^4 + 5x^2 + 3$$
$$f(-x) = (-x)^4 + 5(-x)^2 + 3$$
$$f(-x) = x^4 + 5x^2 + 3 = f(x)$$

Question no. 7

Prove that $f(x) = x^3 - 8x$ is an odd function

Solution:

$$f(x) = x^3 - 8x$$
$$f(-x) = (-x)^3 - 8(-x)$$
$$f(-x) = -x^3 + 8x = -(x^3 - 8x) = -f(x)$$

Question no. 8

Express the quadratic polynomial $x^2 + 4x + 6$ in complete square form.

Solution:

$x^2 + 4x + 6 = x^2 + 4x + 4 - 4 + 6 = (x+2)^2 + 2$

Question no. 9

If the functions f is defined by

$$f(x) = \frac{1}{x}, \ 0 < x \leq 5,$$

then find the range of f.

Solution:

Since $x \leq 5$, $1/x \geq 1/5$.. i.e., $f(x) \geq 1/5$.

Question no. 10

If the functions f and g are defined by

$$f(x) = \frac{3}{x}, \qquad 0 < x \leq 3,$$
$$g(x) = 5x - 1, \quad x \in R,$$

then Calculate $gf(2)$.

Solution:

$gf(x) = g(f(x)) = g(3/x) = 5(3/x) - 1 = (15/x) - 1.$

So, $gf(2) = (15/2) - 1 = 13/2$

Question no. 11

If the functions g is defined by

$$g(x) = 8x - 5, \quad x \in R,$$

then find an expression in terms of x for $g^{-1}(x)$.

Solution:

To find $g^{-1}(x)$, we write $y = 8x - 5$

$\rightarrow y + 5 = 8x \rightarrow x = (y + 5)/8.$

Therefore, $g^{-1}(x) = (x + 5)/8$

Question no. 12

Let $f(x) = 6x + 1$ and $g(x) = -9x + 5$. Find the product of f & g *i.e* $(f.g)(x)$.

Solution:

$$(f.g)(x) = f(x).g(x) = (6x+1).(-9x+5)$$
$$= -54x^2 + 21x + 5 \ (Ans.)$$